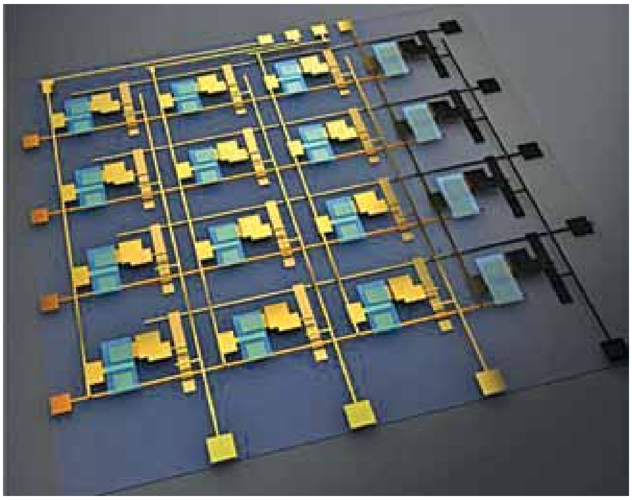
Integration of electrochemical RAM onto silicon transistors achieved

Researchers at University of Illinois Urbana-Champaign have achieved the first material-level integration of electrochemical random-access memory, ECRAM, onto silicon transistors. ECRAM encodes data via ion shuffling be-



ECRAM array (Credit: The Grainger College of Engineering at University of Illinois Urbana-Champaign)

tween gate and channel, with electrical pulses controlling ion movement. The resulting conductivity change stores information, read by measuring current flow. An electrolyte prevents ion flow, making ECRAM non-volatile. Standard microelectronics control devices made of tungsten oxide, zirconium oxide, and mobile protons. These materials are compatible with silicon microfabrication techniques. The use of the same material for gate and channel simplifies control and increases reliability. Protons enable rapid switching, and the channel reliably holds ions for hours. Devices lasted over 100 million cycles and were more efficient than standard memory. Microfabrication-compatible materials allow for shrinking to micro/nanoscale.

Temperature-resistant artificial skin with antibacterial properties developed

Researchers from Dalian University of Technology have designed a new highly flexible and temperature-resistant artificial skin that also possesses antibacterial properties. The skin is a photonic-ionic system that emits optical and electrical signals. Monoglyceride laurate destroys 99.9% of bacteria and fungus. External stimuli alter the photonic nanostructure, changing colour and ion transport, enabling dual-signal response. The skin's monoglyceride laurate can eliminate most gram-positive bacteria and fungi, making it useful for medical technologies like prostheses. The researchers found that its synchronous output of two signals even allowed it to distinguish between different environmental and tactile stimuli.

Repair technique allows small aerial robots to fly despite damaged artificial muscles

MIT researchers have developed repair techniques that allow a bug-sized aerial robot to fly effectively despite severe damage to its artificial muscles. They improved the muscles to isolate defects and recover from minor damage like small holes. They also tested a laser repair method for major damage like fire. Their techniques allowed a robot to

The insect-scale aerial robot and its actuators (Credit: Photography and art by Julian Kamzol and Sampson Wilcox)

maintain flight performance despite severe damage, including 10 needles in one artificial muscle, a burnt hole in another, and a 20% cut in the wing tip. Dielectric elastomer actuators (DEAs) power the wings of a flying robot using mechanical forces. These soft artificial muscles consist of layers of elastomer between thin electrodes. Self-clearing can prevent DEA failures caused by tiny defects by disconnecting the local electrode around the defect. High voltage isolates the failure, and the artificial muscle still works.

Detecting hardware Trojans in microchips being explored

Researchers at Ruhr University Bochum, Germany, and the Max Planck Institute for Security and Privacy (MPI-SP) in